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Wriston Manufacturing Corporation

In early 1992, Richard Sullivan, recently appointed vice president in the Heavy Equipment Division (HED) of the Automotive Supplier Group of the Wriston Manufacturing Corporation, scrutinized one more time the P&L forecast for the Detroit plant—part of a lengthy report on the future of the plant which had been prepared by a task force Sullivan had appointed six months earlier. Sullivan had joined the division in 1988 as division controller, and for several years had watched the plant perform at a level well below division expectations. In addition, he sensed that the plant had lost its spirit. Over the years, products with growth potential had been transferred to new plants and with them had gone investment dollars and management talent. "For the past 20 years," a plant engineer said, "people have been expecting the plant to close."

The Detroit plant's sales were expected to remain in the $35-40 million range, and the task force had concluded that "at best a break-even operation is expected for at least five years if the operation is left as is." Sullivan noted, "On the first cut, it looks like we cannot achieve an acceptable level of profitability at Detroit even if we raised prices and cut hourly wages." With the Detroit facility now his direct responsibility, Sullivan felt it important to address the problem. He did not believe that the existing plant was viable in the long run; in the shorter run, however, he saw three major alternatives:

1. Close the plant as soon as possible and transfer its products to other plants.

2. Invest in plant tooling in an attempt to develop a viable operation for at least the next 5- to 10-year period.

   投资工厂工具，试图在至少未来5到10年的时间里开发一个可行的操作

3. Build a new plant.

Sullivan knew that the decision would be difficult. "We must," he noted, "consider the company's responsibilities to the employees and to our customers; and we must look beyond the Detroit plant to the needs of the division and the corporation. In the end, however, the decision may well boil down to what we think we can do with the people we have available."

Heavy Equipment Division

车轴和制动器

The Heavy Equipment Division (HED) was a large axle and brake manufacturer—one of many divisions in a multi-billion dollar corporation whose products were targeted primarily at the North American transportation industry (see **Exhibit 1** for a typical HED product). Wriston's 1990 annual report stated:

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During 1990, the Heavy Equipment Division continued to lead the original equipment field through the marketing of improved products. These included axles for heavy-duty vehicles, both on- and off-highway; a new family of front-drive axles for 4- and 6-wheel smaller trucks; a new line of non-drive front axles; and an improved hydraulic brake for the truck market. Improved products were also introduced in the off-highway market for construction, agricultural, and mining equipment. In addition, the Heavy Equipment Division continues to manufacture drive train and suspension components for several U.S. Army vehicles.

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Wriston's sales had been declining for three years, led by a precipitous decline in its other divisions. Divisions such as HED were therefore under pressure to perform well, and investment proposals were scrutinized carefully. (HED accounted for approximately 70% of the Automotive Supplier Group's sales in 1991. The Automotive Supplier Group accounted for 16% of Wriston's total revenues, and 28% of its total income.)

HED was headquartered in Detroit. In 1991 the division had nine plants on-stream and a tenth under construction (see **Exhibits 2A** and **2B** for plant data). One senior HED manager who had managed several of these facilities observed, "It's funny how these plants each develop their own unique character—sort of how your children grow up with unique characters, even though they're a part of the same family. Some plants are a joy to live in—things just hum, day in and day out. And others are incredibly complex, like incorrigible problem children—as soon as you get one problem solved, you've got another to deal with."

three levels to measure product complecity: product line, product families, produce model.

The complexity of the product missions assigned to each plant was, in fact, quite different. Experienced manufacturers at HED tended to think of the complexity of a plant manager's job in terms of the range of products his or her plant produced. Complexity could be measured at three levels. The **product line** was the broadest measure of mission complexity, and manufacturing at HED was allocated at its highest level according to its three major product lines: axles for on-highway vehicles (mainly trucks), axles for off-highway applications (such as construction, mining, and agricultural equipment), and brakes. For a number of years, the division had been moving toward plant specialization by product line. "The reason for this," Sullivan explained, "was to minimize fixed investment in machine tools by tooling for a product line in only one location." He added, "it's also easier to manage a plant that's more focused."

The on-highway axles were currently produced in five different plants: Detroit, MI, Lebanon, PA, Tiffin, OH, Fremont, OH, and Maysville, KY; in addition, a new on-highway axle plant in Lancaster, OH would soon come on-line. Off-highway axles were made in Detroit, MI, Saginaw, MI, and Lima, OH. Brakes were made in three plants—Sandusky, OH made components and assembled brakes for the U.S. market; an Essex, Canada plant made brakes for the Canadian market; and the Detroit plant made replacement parts for old-model brakes.

The next level at which complexity could be measured was in terms of **product families**. Different product families typically required different routings through a plant's component manufacturing and assembly areas. For example, "drive axles"[1] which transmitted power from the drive shaft to the wheels had many more components, and required many more assembly operations, than did "non-drive" (generally front) axles. Hence, the process flows for these two classes of products were very different. Even within the drive axle category, there were many different product families, each requiring a substantially different manufacturing process flow.

Simple non-drive on-highway front axles were all made in Lebanon, PA. Families of the more complex rear axles were assigned to the remaining five on-highway plants primarily on the

[1] A *drive* axle is an axle that transmits power from the engine to the wheels on the axle. For example, in a vehicle for which power from the engine is transmitted via the drive shaft to the rear axle, the rear axle is the *drive* axle, and the front axle would be termed a *non-drive* axle.

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basis of volume. The Detroit plant made most of the "low runner" (low volume) families including prototypes for new products and replacement parts for old products. The Tiffin, Ohio, plant was a feeder plant for the medium-volume Fremont, Ohio, plant and the high-volume Maysville, Kentucky, plant. The Tiffin plant manufactured components that were common to the axles produced in both plants. The Fremont and Maysville plants manufactured components that were unique to their particular product lines and assembled the axles. With the completion of the Lancaster, Ohio, plant, the task of assembling the higher-volume families in the Fremont product mix would be transferred to Lancaster. Tiffin would remain a feeder plant providing Fremont, Maysville, and Lancaster with common components. Fremont would continue both to manufacture components for use at Fremont and Lancaster, and to assemble lower-volume families. Richard Sullivan explained the Lancaster decision:

> A key in our business is the technology of gear machining. Fremont is a very high technology plant, our strongest in the gear machining area. We needed to expand that machining capacity, but felt that the Fremont plant was already as large a plant as we wanted. So we decided that it would be less expensive to build an assembly plant nearby in Lancaster and let Fremont focus mainly on machining, than it would be to build a new, integrated plant to do both.

The third level at which the complexity of a plant's mission was measured was the **product model**. A product family could consist of a single model, or as many as 15 to 20 models. Models within a family followed the same routing through a factory, but often required different tooling and set-ups in component machining operations, or different fixtures and tools in assembly operations (reflecting differences in size or features across models within the product family). Often the variations between models in a given family occurred at later stages in the manufacturing process flow, where customer-specific features were built into the products. **Exhibit 2A** details the numbers of product lines, families, and models manufactured in each of the Heavy Equipment Division's plants in 1991.

Each of the plants was expected to stand alone in terms of profitability. Plant managers were evaluated on a variety of measures, but return on assets employed (net fixed assets plus working capital) was a major concern. **Exhibit 2B** describes the 1991 financial performance of each of the plants in the HED system.

Division Concerns

Upon taking charge at HED, Sullivan found managing the division's complex set of manufacturing facilities to be among his most challenging problems, because the financial performance varied widely. For example, the Maysville plant was finally beginning to perform reasonably well. The first part had been machined there in 1988. The plant had lost money in 1988 and 1989 but had shown a small profit in 1990. Sullivan worried that "sufficient management quality and depth was the single most difficult challenge in the Maysville start-up, and now we have Lancaster coming on-stream."

In addition, the Lima, Saginaw, and Detroit plants presented problems in 1991. The capital-intensive Lima plant, designed for high volume products, had become a problem area the previous four years. It had been built for a sales level of $60 million; however, volume had declined to about $12 million in 1990, with 70% accounted for by one customer. In October 1990 virtually the entire management team had been eliminated; since then, the plant had operated essentially as a department for Saginaw. Sullivan was considering either selling the plant and machinery to the major customer or keeping the plant open and transferring in products from Saginaw. In contrast, on a 2.5 shift, 5 days/week basis, Saginaw ran at 94% of capacity during 1991.

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A third concern for Sullivan was HED's product line, which seemed to have expanded inexorably for the past two decades. New models needed to be added each year to keep pace with customers' new product introductions, but it was difficult to delete models from the line. Customers' products—trucks, tractors, and construction equipment—had long lives: even after customers no longer purchased a given model, the demand for replacement parts often continued for 20 years or more.

The Detroit Plant

The Detroit plant, built in 1914, was the division's first plant, located near division headquarters. Almost all products of the division had roots that could be traced to Detroit. Over the years, as the market for a new product grew, production had been transferred to a new plant specifically built or acquired for it.

In 1947 brakes had been transferred from Detroit to Sandusky. One of the higher-volume on-highway rear axles was moved to Tiffin in 1952. When Tiffin was dedicated to component manufacture, assembly of the axle line was transferred to Fremont. Some on-highway products had developed into very high-volume standardized products used by several original equipment manufacturers and eventually were switched from Fremont to Maysville.

so detroit produces the residual of high-volume, specified products, that are low-volume, multi-sector products.

The Detroit plant had thus been left with a residue of low-volume products, and, unlike other plants in the division, with two product lines: 60% of sales were on-highway axles, 40% off-highway axles. In addition, it retained primary responsibility for "service" (replacement) parts for all three of the division's product lines.[2] Detroit plant customers ranged from large manufacturers to small operators totally dependent upon the company for their axles. (See **Exhibit 3** for information regarding the Detroit plant's financial performance.)

The Detroit plant manager was 58-year-old Frank Kravitz, a long-time HED employee who had started his career as a supervisor at the Detroit plant. Kravitz had left Detroit to accompany a product transfer to Tiffin and had moved later to Fremont. He had returned to Detroit as plant manager in the late seventies. A prolonged illness had kept him away from work for a significant part of 1991.

Based upon his experience as division controller, Sullivan knew that several factors (most resulting from the pattern of transferring products out of the plant once they reached reasonably high volume levels) had contributed to the poor performance of the Detroit plant.

Investment. The plants to which Detroit transferred growth/high-volume products were usually very successful, achieving high returns on assets employed. Their managers became heroes within a division that treated plants as profit centers. Over time, available investment dollars flowed to the newer plants, because to increase overall return on assets, capital in the corporation tended to be allocated to those products and facilities with the highest demonstrable rates of return. (Wriston used a 10% after tax hurdle rate to evaluate investment options.) The Detroit plant could not compete effectively for these funds because capital was seldom allocated to the low-volume, and often dying, products that remained there. The condition of the machine tools—the heart of the plant—reflected the lack of investment. (See **Exhibit 4** for investment and depreciation trends at the Detroit plant.)

Machine tools. These large machines removed metal from rough forgings and castings to produce the exact shape and size specified for the finished parts. A variety of machine types was used—drilling, boring, grinding, milling, turning. Each had jigs in which parts were positioned securely, and cutters were used to remove the metal. While the plant did have four numerically

[2]The division was committed to providing replacement parts for all of its products, including out-of-production models.

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controlled machines (modern tools that ran automatically), many of the tools were "antiques," including some still driven by wide leather belts from an overhead power shaft. Because of the low volumes per model characteristic of products made in Detroit, with the current batch sizes, set-up times could easily run 10 times the run time for a batch of parts.

The machines were generally grouped on the plant floor by type of machine. Sullivan explained, "keeping the layout organized by machining department keeps us flexible, since our products and the processes change over time. It also keeps our training costs low, since operators can specialize on one type of machine." Products moved from one department to another according to the machining and assembly sequence specified on a computer-generated routing sheet which accompanied each batch of parts.

The average age of the machine tools in the Detroit axle plant was 33.1 years; in the Heavy Equipment Division, 15.9 years.

The plant. The Detroit plant was a collection of 12 buildings that had been added upon and cobbled together in a piecemeal, unplanned fashion between 1914 and 1960. A 1990 report had stated: "The plant has degenerated to the point that major improvements will be mandatory in a few years. The electrical system is inadequate. The water system is constantly springing leaks. Column spacing and ceiling heights compromise storage and machine layouts. A major improvement has been made to the fire sprinkler system, but it is still below our insurance underwriters' standards."

Labor. The plant was unionized by the United Automobile Workers of America (UAW), which had a national agreement with Wriston Manufacturing covering all plants in the division. Labor relations were described as good. Machine operators were skilled workers, with expertise both in set-up procedures and in monitoring machines during runs. The plant had a history of long-service employment (see **Exhibit 5** for the age distribution of hourly employees). Many believed, however, that "bad labor habits" had developed in the Detroit plant. The following statement appeared in a local newspaper in September 1991:

Errol Jenkins, Wriston's group vice president of manufacturing, said his Automotive Division has a "very difficult labor force" in Detroit. "Mondays and Fridays are bad days for us. Absenteeism is high. We have high turnover." He said the company instituted programs to get "people used to good work habits" but they haven't paid off and the firm has decided to locate new plants in small communities.

The Detroit Plant study task force had elaborated on this problem. Its report contained the following:

> Absenteeism and turnover are problems that are steadily deteriorating. We cite two main factors:
>
> 1. Hourly employee ages split into two groups: above 50 and below 30. A degree of polarization exists that reduces the effectiveness of the entire group. The lower performance against standards of the younger, newer employees had resulted in lower output from the older employees too.
>
> 2. The culture and expectations of the present work force have changed markedly. The desire to strive for the "better things in life" and for promotion is not the same as it used to be. Wages are high compared to what the new employees are used to. Newer employees seem to feel it isn't necessary to work every scheduled day. Nor do they seem to have much loyalty to Wriston Manufacturing—they can be trained today, and gone tomorrow.

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In November 1991, Sullivan received a letter from the president of the UAW local with jurisdiction for the Detroit plant. The letter is reproduced in **Exhibit 6**.

Sullivan also suspected, however, that with little plant investment or attention, Detroit workers believed corporate expectations for them were low and had begun to perform accordingly. The Detroit plant overhead was significantly higher than that of other division plants. In part, Sullivan felt, this was due to the increasing maintenance costs associated with the old facility and machines.

底特律工厂的管理费用明显高于其他分公司的工厂

Richard Sullivan felt a strong obligation to his employees, past and present:

> If we decide to transfer more work out of Detroit, we should offer our senior employees the opportunity to transfer to another one of our plants or to elect early retirement. We owe it to these people who really have helped build up the division over the years. It has been management's decision to transfer high-volume profitable products out of Detroit to the other plants.
>
> We are not contractually obligated to offer job security benefits to our younger employees. However, the trend is for unions to bargain for job security clauses. To date, these clauses have been used mainly as bargaining chips that have been given up in actual negotiations. Recently, however, unions have begun to press for this item. If we wait too much longer in closing Detroit, we may have contractual obligations to provide job security for all our employees. Right now our only obligation for job security is a moral one.

Product costs. All plants in the division used a uniform product costing system, which calculated a standard cost for each model produced. Materials costs for each model were taken directly from the bill of purchased materials required for each part. Cost added in manufacturing was calculated by multiplying the standard number of labor hours required to make the components and assemble the model, by the "fully costed" hourly labor rate (i.e the hourly wage rate plus an hourly variable burden). The hourly variable burden was calculated as the ratio of variable manufacturing overheads in each plant to that plant's direct labor cost (the "variable burden rate") and multiplying that by the average hourly wage of the direct labor force. Product cost was therefore the sum of materials cost, direct labor, and an allocation of variable overhead cost per unit. Variable overhead costs included first-line supervisors' wages, costs of set-up labor, scrap and rework costs, and fringe benefit costs for all direct and variable overhead laborers. Gross margins generated per model were calculated by subtracting product cost from the sales price. "One reason we've repeatedly had to transfer higher volume products out of Detroit is that it seems that we can make them for a lot less in the other, more modern plants," Sullivan noted.

On occasion, Division staff attempted to show a more fully allocated picture of product line profitability by allocating fixed plant overhead to products, on a percent-of-sales basis. Fixed overhead included depreciation, utilities, and the salaries and fringe benefit costs of employees involved in general management, accounting, personnel, purchasing, production scheduling, expediting, materials handling, materials control, maintenance, etc.

Although calculating full product costs for each model was a cumbersome process, division management found the detailed information valuable in establishing prices on the division's wide array of products.

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The Detroit Plant Study Group

Sullivan had requested a feasibility study in August 1991 to identify and evaluate alternatives for Detroit. He had received the report in mid-January 1992; had circulated it to each of his plant managers and key staff members for reactions; and was working to formulate a plan of action for the Detroit Plant. Sullivan noted that the study team had put the Detroit Plant's products in three general groups: on-highway axles that are economically worth continuing to produce; off-highway axles that are worth continuing; and both on- and off-highway axles that are not economically justified. (See **Exhibit 7** for excerpts from that report.)

About a week after he distributed the study groups recommendations, Sullivan got a phone call from Frank Kravitz at the Detroit Plant. "Look Richard," Kravitz began, "I got this report. I'm not calling to argue about their numbers, because in a lot of ways they make sense. I mean, we've been waffling on this for a long time, and maybe we really should walk away from this place. But there's something I can't figure out. I know our overheads are bad. But I walk around here and ask, "Where's the overhead? I mean, we haven't painted *anything* here in 20 years. The bathroom fixtures are at least 40 years old. We closed the employee cafeteria five years ago. We have one secretary in the whole plant, and she triples as receptionist and telephone operator. On the other hand, when I go to the other plants, like Lebanon, Fremont, and Maysville, I can *see* the overhead cost. They have nice cafeterias, wooden desks in their offices, receptionists and secretaries, and so on. The only place you *can't* see their overheads is in the burden rates. I just don't understand it."

After Kravitz's call, Sullivan flipped through the study group's recommendations again. "I don't understand it either, but it's there. We've tried and tried, but the Detroit plant is just a high-cost facility. Based on these numbers, we just can't justify staying in Detroit. We'd make a *lot* more money on that first category of Detroit products if we made them in a lower-cost plant. But one thing that intrigues me," he thought to himself, "is that there has been talk of shutting down the plant for years, but it never seems to happen. A major consideration, of course, has been the employees. I expect the termination costs would approach $6 million. And this has been a bleak fall in the city of Detroit—the media now greets word of another plant closing with phrases such as 'another Detroit company bit the dust today.' A further consideration is our customers. If we close Detroit, where do we transfer the better products? Should we build a new plant for the division for low-volume products? I worry that discontinuing products might be viewed as arbitrary and drive some customers to seek alternative sources for products they now purchase from other plants in the division. We also don't want to give competitors any openings in the heavy-axle market. The other thing to consider is that Wriston's profit and cash flow situation are pretty tight right now. Maybe it would make sense to keep operating Detroit for a few more years, while we make longer-range plans. In this case, are there any feasible short-term initiatives we could tackle to make the place more profitable?

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Exhibit 1 A Wriston Manufacturing Corporation On-Highway Axle

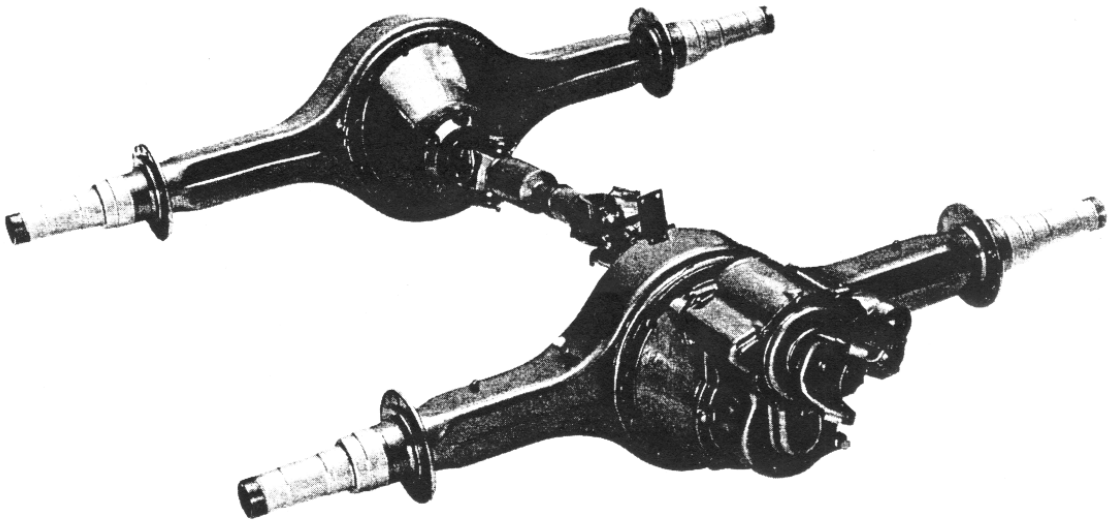

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Exhibit 2A Heavy Equipment Division Plant Network (as of January 1992)

Plant	Product Lines	1991 Sales (millions of dollars)	Employees	Capacity (millions of dollars)	Return on Assets (%)	Variable Burden Rate[1]	Total Burden Rate[2]	Number of Product Families[3]	Number of Product Models[3]
Sandusky, OH	US brakes	$ 66.8	467	$100	29	1.11	3.58	5	10
Essex, Canada	Canada brakes	22.6	197	30	18	1.64	5.30	4	8
Detroit, MI	Low volume rear on-highway axles, Low volume off-highway axles, & Replacement parts	34.8	352	50	-7	2.13	6.00	20	120
Saginaw, MI	Medium volume off-highway axles	94.2	758	100	48	1.35	4.10	10	110
Lima, OH	High volume off-highway axles	12.0	124	60	-12	1.56	5.05	4	80
Lebanon, PA	Simple front on-highway axles	72.6	500	90	37	0.85	2.64	2	30
Tiffin, OH	Rear on-highway axle components	91.6	554	120	30	1.12	3.50	6	60
Fremont, OH	Medium volume rear on-highway axles	158.0	1,266	180	36	1.20	3.65	10	40
Lancaster, OH	High volume rear on-highway axles		1,000[4]	106[4]					
Maysville, KY	High volume rear on-highway axles	110.0	807	150	26	0.71	2.35	2	20

[1] Defined as variable manufacturing overhead, divided by direct labor cost.
[2] Defined as variable plus all other manufacturing overhead costs, divided by direct labor cost.
[3] The numbers for the Detroit plant do not include discontinued parts and parts made as replacements by special customer order.
[4] Preliminary estimates.

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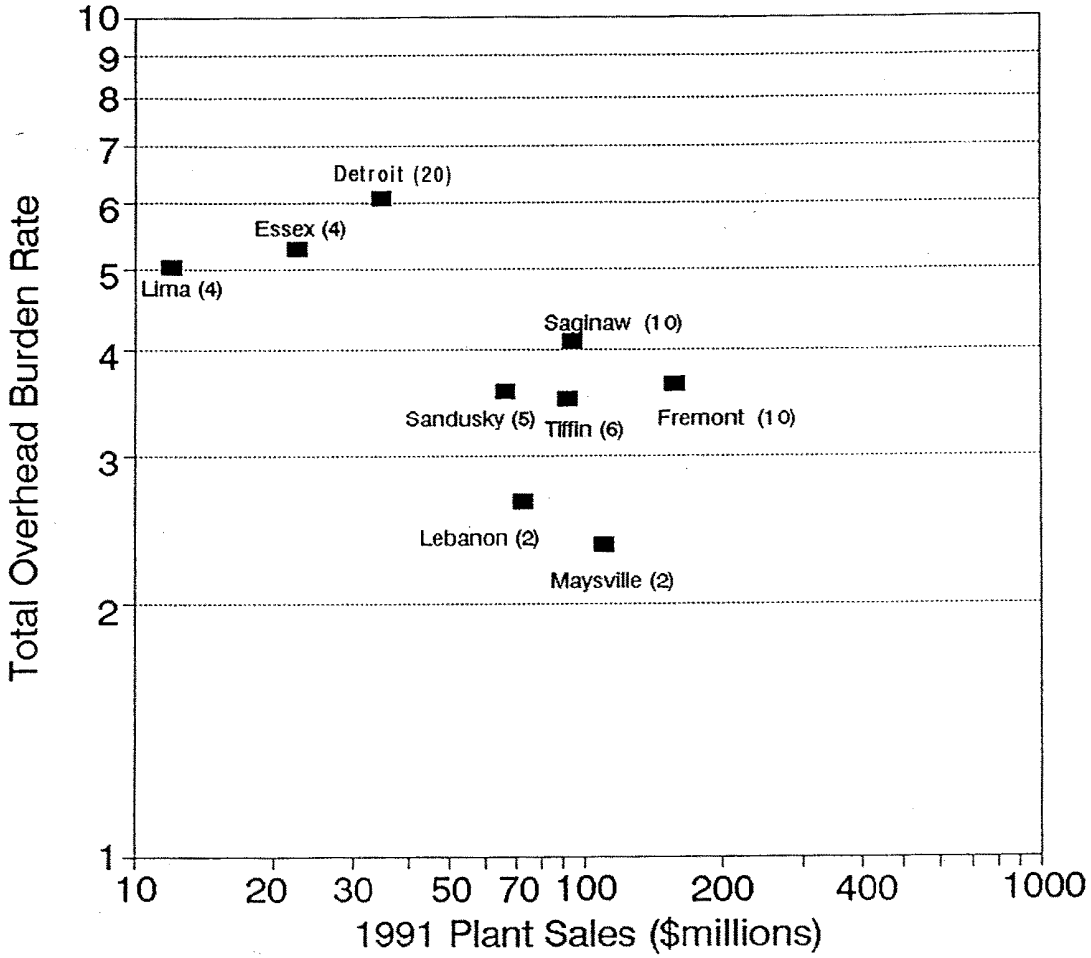
Exhibit 2B The Relationship between Total Plant Sales and Overhead Burden Rates in Wriston's Plant Network



Note: The numbers in parentheses next to each plant's name represent the number of product families manufactured in each plant. Data used to construct this chart were taken directly from **Exhibit 2A**. Both axes have been plotted on a logarithmic scale.

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Exhibit 3 Detroit Plant Financial Performance, 1988-1991 (thousands of dollars)

Income Statement	1991	1990	1989	1988
Sales	$34,720	$40,456	$43,150	$35,726
Cost of Sales:				
Direct Labor	2,738	3,220	3,280	3,174
Materials	16,550	18,610	18,986	16,254
Variable Mfg. Overhead	5,832	6,794	6,822	6,208
Cost of Goods Sold	25,120	28,624	29,088	25,636
Gross Profit	9,600	11,830	14,602	10,090
Other overhead costs	10,528	13,190	17,398	12,300
Contribution to Division	($928)	($1,360)	($2,796)	($2,210)
Assets				
Cash	$190	$230	$220	$160
Receivables	4,544	4,172	5,716	4,450
Inventory	18,840	20,414	19,828	15,046
Net Fixed Assets	3,940	4,520	5,200	5,960
Total Assets	$27,514	$29,336	$30,964	$25,616

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Exhibit 4 Detroit Plant Investment and Depreciation Trends, 1980-1990

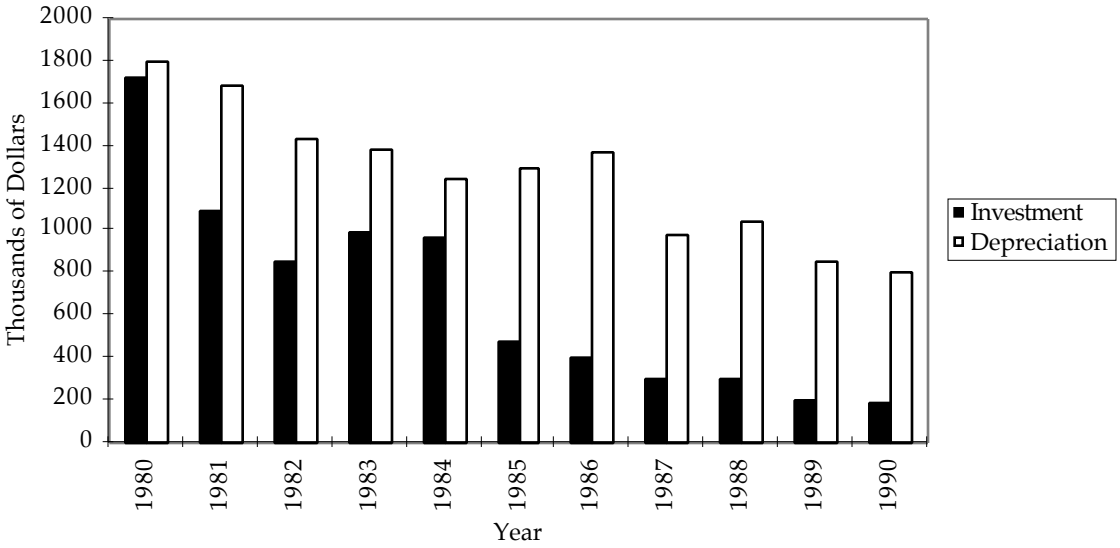

Exhibit 5 Age Profile of Detroit Plant Hourly Employees, 1991

(Total number of workers = 352)

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Exhibit 6 UAW Letter to Richard Sullivan, November 11, 1991

Dear Mr. Sullivan:

We write to you asking for answers to several questions which we believe are important to the future of the Detroit plant.

The questions are based on the following:

1. The many statements made by management about the plant's competitive position.

2. Of most concern, the statements allegedly made by Mr. Jenkins which were directly quoted in a September newsletter article.

The union believes there are insinuations in Mr. Jenkins' statement which underline the need for clarification as soon as possible concerning possible plant closings.

We would appreciate your immediate response regarding these very important questions.

1. What is the purpose of the "feasibility study" of the Detroit plant; what does this study consist of, and will the union be offered the opportunity to see the results?

2. When and if this study is completed will the union members be notified of its results and will they have the opportunity to voice their opinions before any official decision is made about the plant's future?

We believe the employees should know, and have a right to know, what the outlook is for the plant's future. We would like to repeat our suggestion that the company take a look at the feasibility of the four-day work week which could help resolve the problem of absenteeism on Fridays and Mondays.

The union believes that the answers to these questions will give the workers an understanding of the company's problems and some possible resolutions in an attempt to solidify the workers' job security.

In answering these questions, we hope you will help us understand the problems with which we are both faced.

Very truly yours,

John Carter
President, Local

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Exhibit 7 Excerpts from the Report of the Detroit Plant Study Group

1991 Financial Results for the Three Product Groupings at Detroit ($000)

	Group 1	Group 2	Group 3	Total
Sales	$ 13,888	$ 9,258	$ 11,574	$ 34,720
Direct Labor	$ 592	$ 516	$ 1,630	$ 2,738
Materials	$ 6,390	$ 4,340	$ 5,820	$ 16,550
Variable Mfg. Overhead	$ 1,260	$ 1,102	$ 3,470	$ 5,832
Fixed Mfg. Overhead	$ 3,829	$ 2,582	$ 4,116	$ 10,528
Profit (Loss)	$ 1,816	$ 717	$ (3,462)	$ (928)

where:

Group 1: About two-thirds of the on-highway axles. These look like they could economically be continued;

Group 2: About two-thirds of the off-highway axles. These also look like they could economically be continued;

Group 3: The remainder, both on-highway and off-highway. We believe these cannot economically be produced.

Comments about Group 1

We could transfer these products to Fremont, Maysville, or Lancaster. We estimate an average direct labor savings of 5% due to the better machines and more productive work forces in those plants. We also estimate a materials cost savings of 2% due to lower scrap rates and higher-volume purchasing in those plants. The biggest improvement in the profitability of these products comes from overhead savings, however—as you know, the burden rates in these plants are around half of the Detroit rate. If we move these products, we will incur one-time tooling costs of about $17 million. We could take much better advantage of economies of scale in some of our under-utilized plants if we loaded them more fully with volume. Costs should drop across the board—on the Detroit products we transfer in, and on the products already there. Overall, taking all of these financial implications into consideration, we estimate that our annual after-tax cash flow would increase approximately $2.3 million per year for the next 20 years if we move the Group 1 products to Fremont, $2.4 million per year if we move them to Maysville, and $2.7 million per year if we move them to Lancaster.

Comments about Group 2

If we transferred these axles, they'd go to Saginaw or Lima. We estimate a direct labor savings there of 6% due to the better machines and very productive work force, and materials savings of about 1%. The advantage to Saginaw, if it has the capacity to absorb these products, is that it is a lower-burden plant. We would incur a one-time additional tooling for the Detroit products of about $8 million; and overhead costs at Saginaw or Lima would increase somewhat. Overall, we estimate that our annual after-tax cash flow would increase approximately $1 million per year for the next twenty years if we move the Group 2 products to Saginaw and $0.7 million if we move them to Lima.

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Comments on Group 3

Although we could almost produce these products in *any* other plant at lower costs than we do in Detroit, there doesn't seem to be much hope for economic production of these axles; their direct labor content is too high, and volume is just too low. The question is: if we drop the axles, what do we do for our customers? Overall, we estimate that our annual after-tax cash flows would increase approximately $1.9 million per year if we dropped the Group 3 products.

Alternatives for the Detroit Plant

Closing Detroit could bring in $4 million from sale of the property. We'd also incur employee termination costs of about $6 million. If we stayed in Detroit, maintaining the facility and buying a few necessary tools would probably cost $2 million per year. We could probably keep Detroit running at its current level of profitability another 5 to 10 years before it completely falls apart. Then we'd have to decide whether to build a new plant or close down the Detroit plant.

A New Plant

If we were to build a new low-volume plant for the division, we could probably increase our annual cash flows (after accounting for plant depreciation, etc.) by about $3 million per year. We'd have to invest about $30 million in the plant and tooling, and start-up costs would cost another $6 million. We'd receive $4 million from the sale of the Detroit plant property, but would not incur employee termination costs.

Note: Wriston used a 10% after-tax hurdle rate to evaluate investment options.

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